Chomsky Normal form

In a CFG, there is no restriction on the RHS of a production. The restrictions are imposed on the RHS of productions in a CFG resulting in various normal forms.

The different normal forms are

1. Chomsky Normal form (CNF)
2. Greibach Normal form (GNF)

Def^n : Let $G=(V,T,P,S)$ be a CFG. The grammar G is said to be in CNF if all productions are of the form $A \to BC$ or $A \to a$ where $A,B,C \in V$ and $a \in T$.

Theorem :

Let $G=(V,T,P,S)$ be a CFG which generates context free language without ε. find an equivalent context free grammar $G'=(V',T,P',S)$ in CNF such that $L(G)=L(G')$

Proof :

Let the grammar G has no ε-productions and unit productions. The grammar G' can be obtained using the following steps.

STEP 1 : Consider the productions of the form

$A \to X_1X_2X_3 \dots X_n$ where $n \geq 2$ and each $X_i \in (V \cup T)$.

* If X is a terminal then replace this by a corresponding non-terminal B_a and introduce the production $B_a \to a$.
* The non-terminals on the RHS of the production are retained.
* The resulting productions are added to P_1. The resulting CFG $G_1=(V_1,T_1,P_1,S)$ where each production in P_1 is of the form $A \to A_1A_2 \dots A_n$ or $A \to a$ generates the same language as accepted by grammar G. So, $L(G)=L(G_1)$.

STEP 2 : Restrict the number of variables on the RHS of the production.

Add all the productions of G_1 which are in CNF to P'.

Consider a production of the form

$A \to A_1A_2 \dots A_n$ where $n \geq 3$. The production can be written as,

$A \to A_1D_1$

$D_1 \to A_2D_2$

$D_2 \to A_3D_3$

$\vdots$

$D_{n-2} \to A_{n-1}D_{n-1}$

These productions are added to P' and new variables are added to V'. The resulting grammar $G'=(V',T,P',S)$ generates the same language as accepted by G i.e, $L(G)=L(G')$.

Q1. Consider the grammar

$S \rightarrow 0A \mid 1B$

$A \rightarrow 0AA \mid 1S \mid 1$

$B \rightarrow 1BB \mid 0S \mid 0$ Obtain the Grammar in CNF.

The productions which are in CNF are added to P_1.

i.e., $A \rightarrow 1$

$B \rightarrow 0$

STEP 1 :

Given productions	Action	Resulting productions
$S \rightarrow 0A \mid 1B$	Replace 0 by B_0 and introduce the production $B_0 \rightarrow 0$. Replace 1 by B_1 and introduce the production $B_1 \rightarrow 1$	$S \rightarrow B_0A \mid B_1B$ $B_0 \rightarrow 0$ $B_1 \rightarrow 1$
$A \rightarrow 0AA \mid 1S$	$B_0 \rightarrow 0$ $B_1 \rightarrow 1$	$A \rightarrow B_0AA \mid B_1S$ $B_0 \rightarrow 0$ $B_1 \rightarrow 1$
$B \rightarrow 1BB \mid 0S$	$B_0 \rightarrow 0$ $B_1 \rightarrow 1$	$B \rightarrow B_1BB \mid B_0S$ $B_0 \rightarrow 0$ $B_1 \rightarrow 1$

The grammar $G_1 = (V_1, T, P_1, S)$ is

$V_1 = \{S, A, B, B_0, B_1\}$

$T = \{0, 1\}$

$P_1 = \{ S \rightarrow B_0A \mid B_1B$

$A \rightarrow B_0AA \mid B_1S \mid 1$

$B \rightarrow B_1BB \mid B_0S \mid 0$

$B_0 \rightarrow 0$

$B_1 \rightarrow 1$

$\}$

S is the start symbol.

STEP 2 : Restricting the number of variables on the RHS of the production to 2.

Given productions	Action	Resulting productions
$A \to B_0AA$	Replace AA on RHS with variable	$A \to B_0D_1$
	D_1 and introduce the production	$D_1 \to AA$
	$D_1 \to AA$	
$B \to B_1BB$	Replace BB on RHS with variable	$B \to B_1D_2$
	D_2 and introduce the production	$D_2 \to BB$
	$D_2 \to BB$	

The grammar $G' = (V', T, P', S)$ is in CNF where

$V_1 = \{S, A, B, B_0, B_1, D_1, D_2\}$

$T_1 = \{0, 1\}$

$P_1 = \{ S \to B_0A \mid B_1B$

$A \to B_1S \mid 1 \mid B_0D_1$

$B \to B_0S \mid 0 \mid B_1D_2$

$B_0 \to 0$

$B_1 \to 1$

$D_1 \to AA$

$D_2 \to BB$

$\}$

S is the start symbol

Greibach Normal form (GNF)

Def^n: Let $G = (V, T, P, S)$ be a CFG. The CFG is said to be in GNF if all the productions are of the form $A \to a\alpha$ where $a \in T$ & $\alpha \in V^*$.

Theorem: Let $G = (V, T, P, S)$ be a CFG generating the language L without ε. An equivalent grammar G_1 generating the same language exists for which every production is of the form $A \to a\alpha$. where A is a variable, a is a terminal and α is string of zero or more variables.

Proof: Procedure to obtain the grammar in GNF:

STEP 1: Obtain the grammar in CNF.

STEP 2: Rename the non-terminals to $A_1, A_2, A_3, \ldots\ldots$

STEP 3 : Using substitution method, obtain the productions to the form $A_i \rightarrow A_j\alpha$ for $i<j$ where $\alpha \in V^*$. / $\alpha \in (V-\Sigma)^*$

STEP 4 : After substitution, if a grammar has left-recursion, then eliminate left recursion.

STEP 5 : Apply step 3 and/or step 4 more than once to get the grammar in GNF.

Q1 : Convert the following grammar

$S \rightarrow AB1 \mid 0$

$A \rightarrow 00A \mid B$

$B \rightarrow 1A1$ into GNF.

Note : A grammar should not have any unit productions and ε-productions. In case if the grammar has ε-productions and unit productions perform the following operations one after the other

1. Eliminate all ε-productions
2. Eliminate all unit productions
3. Obtain the grammar in CNF
4. finally convert the grammar into GNF.

The grammar does not have any ε-productions but has unit production $A \rightarrow B$

The dependency graph is (A) $\longrightarrow$ (B)

From the dependency graph, $A \overset{*}{\Rightarrow} B$, So, $A \rightarrow 1A1 \mid 00A$

The grammar after eliminating unit productions is,

$S \rightarrow AB1 \mid 0$

$A \rightarrow 00A \mid 1A1$

$B \rightarrow 1A1$

STEP 1 : ① $S \rightarrow ABA_1 \mid 0$

$A \rightarrow A_0A_0A \mid A_1AA_1$

$B \rightarrow A_1AA_1$

$A_1 \rightarrow 1$

$A_0 \rightarrow 0$

② $S \rightarrow AD_1 \mid 0$

$A \rightarrow A_0D_2 \mid A_1D_3$

$B \rightarrow A_1D_3$

$A_1 \rightarrow 1$

$A_0 \rightarrow 0$

$D_1 \rightarrow BA_1$

$D_2 \rightarrow A_0A$

$D_3 \rightarrow AA_1$

STEP 2: Let $S = A_1$, $A = A_2$, $B = A_3$, $A_0 = A_4$, $A_1 = A_5$, $D_1 = A_6$, $D_2 = A_7$, $D_3 = A_8$.

Grammar can be re-written as

$A_1 \to A_2A_6 \mid 0$ ✓ — $1<2$ ✓

$A_2 \to A_4A_7 \mid A_5A_8$ ✓ — $2<4$ ✓

$A_3 \to A_5A_8$ — $3<5$ ✓

$A_i \to A_jx$ $i<j$ form

$A_5 \to 1$ ✓

$A_4 \to 0$ ✓

} GNF

$A_6 \to A_3A_5$ — $6<3$

$A_7 \to A_4A_2$ ✓ — $7<4$

$A_8 \to A_2A_5$ — $8<2$

} Y

STEP 3: $A_4 \to 0$

$A_5 \to 1$ are in GNF.

Consider A_3 production: Substituting for A_5 in A_3 production,

$A_3 \to A_5A_8 \to 1A_8$ it's in GNF.

Consider A_2 production: Since A_4 and A_5 productions are in GNF, substituting these productions in A_2,

$A_2 \to 0A_7 \mid 1A_8$ it's in GNF.

Consider A_7 production: Substituting A_4 in A_7

$A_7 \to A_4A_2 \to 0A_2$

Consider A_1 production: Since A_2 production is in GNF, substituting for A_2 in A_1 production

$A_1 \to A_2A_6 \mid 0 = (0A_7 \mid 1A_8)A_6 \mid 0 = 0A_7A_6 \mid 1A_8A_6 \mid 0$

it's in GNF.

Consider A_6 production: Since A_3 production is in GNF, substituting for A_3 in A_6,

$A_6 \to A_3A_5 = 1A_8A_5$ it's in GNF.

Consider A_8 production: Since A_2 production is in GNF, substituting for A_2 in A_8,

$A_8 \to A_2A_5 = (0A_7 \mid 1A_8)A_5 = 0A_7A_5 \mid 1A_8A_5$ it's in GNF.

Final grammar which is in GNF is $G = (V, T, P, S)$ where

$V = \{A_1, A_2, A_3, A_4, A_5, A_6, A_7, A_8\}$

$T = \{0, 1\}$

P : { $A_1 \to 0A_7A_6 \mid 1A_8A_6 \mid 0$

$A_2 \to 0A_7 \mid 1A_8$

$A_3 \to 1A_8$

$A_4 \to 0$

$A_5 \to 1$

$A_6 \to 1A_8A_5$

$A_7 \to 0A_2$

$A_8 \to 0A_7A_5 \mid 1A_8A_5$

}

A_1 is the start symbol.

Q2. Convert the following grammar $A \to BC$

$B \to CA \mid b$

$C \to AB \mid a$ into GNF.

STEP1 : Given grammar is in CNF.

STEP2 : Let $A = A_1, B = A_2, C = A_3$ and the resulting grammar is

$A_1 \to A_2A_3$

$A_2 \to A_3A_1 \mid b$

$A_3 \to A_1A_2 \mid a$

$1 < 2$
$2 < 3$
$3 < 1$ ✗

First two productions are of the form

$A_i \to A_jx$ for $i < j$

So, consider A_3 production : Substitute A_1 in A_3,

$A_3 \to A_1A_2 \mid a = A_2A_3A_2 \mid a$ $\quad i < j$ $(3 < 2)$

again replace the first A_2 in A_3,

$A_3 \to A_2A_3A_2 \mid a = (A_3A_1 \mid b)\,A_3A_2 \mid a$

$A_3 \Rightarrow A_3A_1A_3A_2 \mid bA_3A_2 \mid a$

$A_3 \rightarrow A_3A_1A_3A_2 \mid bA_3A_2 \mid a$

which is having left recursion.

$A \rightarrow A\alpha_i \mid \beta_i$	Substitution	$A \rightarrow \beta_iA'$, $A' \rightarrow \alpha_iA' \mid \varepsilon$
$A_3 \rightarrow A_3A_1A_3A_2 \mid$	$A = A_3$	$A_3 \rightarrow bA_3A_2A' \mid aA'$
$bA_3A_2 \mid a$	$\alpha_1 = A_1A_3A_2$	$A' \rightarrow A_1A_3A_2A' \mid \varepsilon$
	$\beta_1 = bA_3A_2$	
	$\beta_2 = a$	

After eliminating left recursion the production is

$A_3 \rightarrow bA_3A_2A' \mid aA'$

$A' \rightarrow A_1A_3A_2A' \mid \varepsilon$

Now, Replace ε in A' production, so. $A' \rightarrow A_1A_3A_2A' \mid A_1A_3A_2$

Replace A' with A_4.

$A_3 \rightarrow bA_3A_2A_4 \mid aA_4 \mid bA_3A_2 \mid a$

$A_4 \rightarrow A_1A_3A_2A_4 \mid A_1A_3A_2$

Now all A_3 productions are in GNF.

Consider A_2 production: Substitute A_3 productions in A_2

$A_2 \rightarrow A_3A_1 \mid b = (bA_3A_2A_4 \mid aA_4 \mid bA_3A_2 \mid a)A_1 \mid b$

$A_2 \rightarrow bA_3A_2A_4A_1 \mid aA_4A_1 \mid bA_3A_2A_1 \mid aA_1 \mid b$

now A_2 is in GNF.

Consider A_1 production: Substitute A_2 in A_1,

$A_1 \rightarrow A_2A_3 = (bA_3A_2A_4A_1 \mid aA_4A_1 \mid bA_3A_2A_1 \mid aA_1 \mid b)A_3$

$A_1 \rightarrow bA_3A_2A_4A_1A_3 \mid aA_4A_1A_3 \mid bA_3A_2A_1A_3 \mid aA_1A_3 \mid bA_3$

now A_1 is in GNF.

Consider A_4 production: Substitute A_1 in A_4,

$A_4 \rightarrow A_1A_3A_2A_4 \mid A_1A_3A_2$

$A_4 \rightarrow bA_3A_2A_4A_1A_3A_3A_2A_4 \mid aA_4A_1A_3A_3A_2A_4 \mid bA_3A_2A_1A_3A_3A_2A_4 \mid aA_1A_3A_3A_2A_4 \mid bA_3A_3A_2A_4$

$A_4 \rightarrow bA_3A_2A_4A_1A_3A_3A_2 \mid aA_4A_1A_3A_3A_2 \mid bA_3A_2A_1A_3A_3A_2 \mid aA_1A_3A_3A_2 \mid bA_3A_3A_2$

now A_4 is in GNF.

B.I.T

So, the final grammar in GNF is

$G = (V, T, P, S)$

where

$V = \{A_1, A_2, A_3, A_4\}$

$T = \{a, b\}$

P : {
$A_1 \rightarrow bA_3A_2A_1A_3 \mid aA_1A_3 \mid bA_3A_2A_4A_1A_3 \mid aA_4A_1A_3 \mid bA_3$

$A_2 \rightarrow bA_3A_2A_1 \mid aA_1 \mid bA_3A_2A_4A_1 \mid aA_4A_1 \mid b$

$A_3 \rightarrow bA_3A_2 \mid a \mid bA_3A_2A_4 \mid aA_4$

$A_4 \rightarrow bA_3A_2A_1A_3A_3A_2 \mid aA_1A_3A_3A_2 \mid bA_3A_2A_4A_1A_3A_3A_2 \mid aA_4A_1A_3A_3A_2 \mid bA_3A_3A_2 \mid$
$bA_3A_2A_1A_3A_3A_2A_4 \mid aA_1A_3A_3A_2A_4 \mid bA_3A_2A_4A_1A_3A_3A_2A_4 \mid aA_4A_1A_3A_3A_2A_4 \mid$
$bA_3A_3A_2A_4$

}

A1 is the start symbol.

Q3. Convert the following grammar $S \rightarrow AA \mid 0$, $A \rightarrow SS \mid 1$ into GNF.

STEP 1: Given grammar is in CNF.

STEP 2: Let $S = A_1$, $A = A_2$.

The resulting grammar is $A_1 \rightarrow A_2A_2 \mid 0$ $\quad i < j$ ✓

$A_2 \rightarrow A_1A_1 \mid 1$ $\quad 2 < 1$ ✗

First production is of the form $A_i \rightarrow A_j\alpha$ where $i<j$

So, consider A_2 production: Substitute A_1 in A_2,

$A_2 \rightarrow A_1A_1 \mid 1 = A_2A_2A_1 \mid 0A_1 \mid 1$

A_2 is having left recursion.

$A \rightarrow A\alpha_i \mid \beta_i$	Substitution	$A \rightarrow \beta_iA'$, $A' \rightarrow \alpha_iA' \mid \varepsilon$
$A_2 \rightarrow A_2A_2A_1 \mid 0A_1 \mid 1$	$A = A_2$ $\alpha_1 = A_2A_1$ $\beta_1 = 0A_1$ $\beta_2 = 1$	$A_2 \rightarrow 0A_1A' \mid 1A'$ $A' \rightarrow A_2A_1A' \mid A_2A_1$

Replace A' with A_3,

So, $A_2 \rightarrow 0A_1A_3 \mid 1A_3 \mid 0A_1 \mid 1$

$A_3 \rightarrow A_2A_1A_3 \mid A_2A_1$

now A_2 is in GNF.

Consider A_1 production : Substitute A_2 in A_1

$A_1 \rightarrow A_2A_2 \mid 0 = 0A_1A_2 \mid 1A_2 \mid 0A_1A_3A_2 \mid 1A_3A_2 \mid 0$

now A_1 is in GNF.

Consider A_3 production : Substitute A_2 in A_3.

$A_3 \rightarrow A_2A_1A_3 \mid A_2A_1$

$A_3 \rightarrow 0A_1A_2A_1A_3 \mid 1A_3A_1A_3 \mid 0A_1A_3A_1 \mid 1A_1A_3 \mid 0A_1A_3A_1 \mid 1A_3A_1 \mid 0A_1A_1 \mid 1A_1$

now A_3 is in GNF.

The final grammar in GNF is $G = (V, T, P, S)$ where

$V = \{A_1, A_2, A_3\}$

$T = \{0, 1\}$

$P = \{ A_1 \rightarrow 0A_1A_2 \mid 1A_2 \mid 0A_1A_3A_2 \mid 1A_3A_2 \mid 0$

$A_2 \rightarrow 0A_1 \mid 1 \mid 0A_1A_3 \mid 1A_3$

$A_3 \rightarrow 0A_1A_1 \mid 1A_1 \mid 0A_1A_3A_1 \mid 1A_3A_1 \mid 0A_1A_1A_3 \mid 1A_3A_1 \mid 0A_1A_3A_1A_3 \mid 1A_3A_1A_3$

$\}$

A_1 is the start symbol.

Q4. Convert the following grammar into GNF.

$S \rightarrow abAB$

$A \rightarrow bAB \mid \varepsilon$

$B \rightarrow Aa \mid \varepsilon$

Eliminate ε productions - $S \rightarrow abAB \mid abA \mid abB \mid ab$

$A \rightarrow bAB \mid bA \mid bB \mid b$

$B \rightarrow Aa \mid a$

STEP 1 : Convert to CNF

① $S \rightarrow A_0A_1AB \mid A_0A_1B \mid A_0A_1A \mid A_0A_1$

$A_0 \rightarrow a$

$A_1 \rightarrow b$

$A \rightarrow A_1AB \mid A_1B \mid A_1A \mid b$

$B \rightarrow A_1A_0 \mid a$

② $S \rightarrow A_0D_1 \mid A_0D_2 \mid A_0D_3 \mid A_0A_1$

$D_1 \rightarrow A_1D_4$, $A_0 \rightarrow a$

$D_4 \rightarrow AB$, $A_1 \rightarrow b$

$D_2 \rightarrow A_1B$

$D_3 \rightarrow A_1A$

$A \rightarrow A_1D_4 \mid A_1B \mid A_1A \mid b$

$B \rightarrow A_1A_0 \mid a$

STEP 2 : Let

$S = A_1$ $\quad A_1 \rightarrow A_4A_6 \mid A_4A_7 \mid A_4A_8 \mid A_4A_5$

$A = A_2$ $\quad A_2 \rightarrow A_5A_9 \mid A_5A_3 \mid A_5A_2 \mid b$

$B = A_3$ $\quad A_3 \rightarrow A_5A_4 \mid a$

$A_0 = A_4$ $\quad A_4 \rightarrow a$

$A_1 = A_5$ $\quad A_5 \rightarrow b$

$D_1 = A_6$ $\quad A_6 \rightarrow A_5A_9$

$D_2 = A_7$ $\quad A_7 \rightarrow A_5A_3$

$D_3 = A_8$ $\quad A_8 \rightarrow A_5A_2$

$D_4 = A_9$ $\quad A_9 \rightarrow A_2A_3$

STEP 3 : Consider A_1 production, substitute A_4 in A_1

$A_1 \rightarrow A_4A_6 \mid A_4A_7 \mid A_4A_8 \mid A_4A_5$

$A_1 \rightarrow aA_6 \mid aA_7 \mid aA_8 \mid aA_5$ it's in GNF.

Consider A_2 production, substitute A_5 in A_2

$A_2 \rightarrow A_5A_9 \mid A_5A_3 \mid A_5A_2 \mid b$

$A_2 \rightarrow bA_9 \mid bA_3 \mid bA_2 \mid b$ it's in GNF.

Consider A_3 production, substitute A_5 in A_3

$A_3 \rightarrow A_5A_4 \mid a$

$A_3 \rightarrow bA_4 \mid a$ it's in GNF.

Consider A_6 production, substitute A_5 in A_6

$A_6 \rightarrow A_5A_9$

$A_6 \rightarrow bA_9$ it's in GNF.

Consider A_7 production, substitute A_5 in A_7

$A_7 \rightarrow A_5A_3$

$A_7 \rightarrow bA_3$ it's in GNF.

Consider A_8 production, substitute A_5 in A_8

$A_8 \rightarrow A_5A_2$

$A_8 \rightarrow bA_2$ it's in GNF.

Consider A_9 production, substitute A_2 in A_9

$A_9 \rightarrow A_2A_3$

$A_9 \rightarrow (bA_9 \mid bA_3 \mid bA_2 \mid b)A_3$

$A_9 \rightarrow bA_9A_3 \mid bA_3A_3 \mid bA_2A_3 \mid bA_3$ it's in GNF.

The final grammar in GNF is $G=(V,T,P,S)$ where

$V = \{ A_1, A_2, A_3, A_4, A_5, A_6, A_7, A_8, A_9 \}$

$T = \{ a, b \}$

$P = \{ A_1 \to aA_6 \mid aA_7 \mid aA_8 \mid aA_5$

$A_2 \to bA_9 \mid bA_3 \mid bA_2 \mid b$

$A_3 \to bA_4 \mid a$

$A_4 \to a$

$A_5 \to b$

$A_6 \to bA_9$

$A_7 \to bA_3$

$A_8 \to bA_2$

$A_9 \to bA_9A_3 \mid bA_3A_3 \mid bA_2A_3 \mid bA_3$

$\}$

$S = \{ A_1 \}$ is the start symbol.

CNF

Q2. Convert the following CFG to CNF.

$S \to aB \mid bA$

$A \to a \mid aS \mid bAA$

$B \to b \mid aS \mid aBB$

① $S \to A_0B \mid A_1A$

$A \to a \mid A_0S \mid A_1AA$

$B \to b \mid A_0S \mid A_0BB$

$A_0 \to a$

$A_1 \to b$

② $S \to A_0B \mid A_1A$

$A \to a \mid A_0S \mid A_1D_1$

$B \to b \mid A_0S \mid A_0D_2$

$D_1 \to AA$, $A_0 \to a$

$D_2 \to BB$, $A_1 \to b$

Q3. Convert the following CFG to CNF. $S \to AB \mid a$

① $S \to AB \mid a$

$A \to A_0A_0A_1$

$B \to AA_2$

$A_0 \to a$

$A_1 \to b$

$A_2 \to c$

② $S \to AB \mid a$

$A \to A_0D_1$, $D_1 \to A_0A_1$

$B \to AA_2$

$A_0 \to a$, $A_1 \to b$, $A_2 \to c$

$A \to aab$

$B \to Ac$

Q4. Obtain the following grammar in CNF.

$S \to aBa \mid abba$

$A \to ab \mid AA$ → Useless production - Remove

$B \to aB \mid a$

$S \to aBa \mid abba$

$B \to aB \mid a$

① $S \to A_0BA_0 \mid A_0A_1A_1A_0$

$A \to A_0A_1 \mid AA$ ✗

$B \to A_0B \mid a$

$A_0 \to a$

$A_1 \to b$

② $S \to A_0D_1 \mid A_0D_2$

$D_1 \to BA_0$

$(D_2 \to A_1A_1A_0)$ $D_2 \to A_1D_3$

$D_3 \to A_1A_0$

~~$A \to A_0A_1 \mid AA$~~ ✗

$B \to A_0B \mid a$

$A_0 \to a$

$A_1 \to b$

Q5. Convert the following CFG to CNF.

$S \to aSb \mid ab \mid Aa$

$A \to aab$

① $S \to A_0SA_1 \mid A_0A_1 \mid AA_0$

$A \to A_0A_0A_1$

$A_0 \to a$

$A_1 \to b$

② $S \to A_0D_1 \mid A_0A_1 \mid AA_0$

$D_1 \to SA_1$

$A \to A_0D_2$

$D_2 \to A_0A_1$

$A_0 \to a$

$A_1 \to b$

Q6. Convert the following CFG to CNF

$E \to E + E \mid E * E \mid (E) \mid id$

① $E \to EE_1E$, $E_1 \to +$

$E \to EE_2E$, $E_2 \to *$

$E \to E_3EE_4$, $E_3 \to ($, $E_4 \to)$

$E \to id$

② $E \to ED_1$, $D_1 \to E_1E$, $E_1 \to +$

$E \to ED_2$, $D_2 \to E_2E$, $E_2 \to *$

$E \to E_3D_3$, $D_3 \to EE_4$, $E_4 \to)$

$E \to id$ $E_3 \to ($

(OR)

$E \to ED_1 \mid ED_2 \mid E_3D_3 \mid id$

$D_1 \to E_1E$, $E_1 \to +$

$D_2 \to E_2E$, $E_2 \to *$

$D_3 \to EE_4$ $E_4 \to)$

$E_3 \to ($

Q7. Convert the following grammar to CNF

$S \to ASB \mid \varepsilon$

$B \to SbS \mid A \mid bb$

$A \to aAS \mid a$

Eliminate ε productions $S \to ASB \mid AB$

$B \to A \mid bb \mid SbS \mid bS \mid Sb \mid b$

$A \to aAS \mid a$

① $S \to ASB \mid AB$

$A \to A_0AA_1 \mid a$

$A_0 \to a$

$A_1 \to S$

$B \to A_0AA_1 \mid a \mid A_0A_0 \mid SA_0S \mid A_0S \mid SA_0 \mid b$

$A_0 \to b$

Eliminate unit production $B \to A$

$S \to ASB \mid AB$

$A \to aAS \mid a$

$B \to aAS \mid a \mid bb \mid SbS \mid bS \mid Sb \mid b$

② $S \to AD_1 \mid AB$

$D_1 \to SB$

$A \to A_0D_2 \mid a$

$D_2 \to AA_1$

$A_0 \to a$

$A_1 \to S$

$A_2 \to b$

$B \to A_0D_2 \mid a \mid A_0A_2 \mid SD_3 \mid A_0S \mid SA_2 \mid b$

$D_3 \to A_2S$

Q8. Convert the following CFG to CNF.

$S \to ASB \mid \varepsilon$

$A \to aAS \mid a$

$B \to SbS \mid A \mid bb$

Eliminate ε productions - $S \to ASB \mid AB$

$A \to a \mid aAS \mid aA$

$B \to SbS \mid bS \mid Sb \mid b \mid A \mid bb$

Eliminate unit productions $S \to ASB \mid AB$

$A \to a \mid aA \mid aAS$

$B \to SbS \mid bS \mid Sb \mid b \mid bb \mid a \mid aA \mid aAS$

① $S \to ASB \mid AB$

$A \to a \mid A_0A \mid A_0AS$

$A_0 \to a$

$B \to SA_1S \mid A_1S \mid SA_1 \mid b \mid A_1A_1 \mid A_0 \mid A_0A \mid A_0AS$

$A_1 \to b$

② $S \to AD_1 \mid AB$

$D_1 \to SB$

$A \to a \mid A_0A \mid A_0D_2$

$D_2 \to AS$

$A_0 \to a$

$A_1 \to b$

$B \to SD_3 \mid A_1S \mid SA_1 \mid b \mid A_1A_1 \mid a \mid A_0A \mid A_0D_2$

$D_3 \to A_1S$

Q9. Convert the following grammar to CNF.

$S \to ABa$

$A \to aab$

$B \to Ac$

① $S \to ABA_0$

$A_0 \to a$

$A \to A_0A_0A_1$

$A_1 \to b$

$B \to AA_2$

$A_2 \to c$

② $S \to AD_1$, $D_1 \to BA_0$

$A \to A_0D_2$, $D_2 \to A_0A_1$

$B \to AA_2$, $A_0 \to a$, $A_1 \to b$, $A_2 \to c$

Pumping Lemma for Context Free Languages.

Statement :

Let L be the context free language and is infinite. Let z is sufficiently long string and $z \in L$ so that $|z| \geq n$ where n is some positive integer. If the string z can be decomposed into combinations of strings $z = uvwxy$ such that $|vwx| \leq n$, $|vx| \geq 1$, then $uv^iwx^iy \in L$ for $i = 0, 1, 2, \ldots$

Proof :

According to Pumping Lemma, it is assumed that string $z \in L$ is finite and is context free language. z is string of terminals which is derived by applying series of productions.

Case 1 : To generate a sufficiently long string z, assume that the string is infinite, one or more variables must be recursive and should be applied more than once.

Let us assume that the language is finite and the grammar has a finite number of variable, assume that all variables are useful. and each production has finite length.

The only way to derive sufficiently long string using such productions is that the grammar should have one or more recursive variables.

Assume that no variable is recursive.

Since no variable is recursive, each variable must be defined only in terms of terminals and other variables. Since those variables are also non-recursive, they have to be defined in terms of terminals and other variables and so on.

By applying the productions like this, there are no variables at all :o

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the final derivation and finally we get string of terminals and the generated string is finite. There is a limit on the length of the string that is generated from the start symbol S. This contradicts the assumption that the Language is finite. Therefore, the assumption that one or more variables are non-recursive is incorrect. It means that one or more variables are recursive and hence the proof.

Case 2 :

$z \in L$ implies that the after applying some/all productions some number of times, we get finally string of terminals and the derivation stops.

Let $z \in L$ is sufficiently long string and so the derivation must have involved recursive use of some non-terminal A and the derivations must have the form $S \overset{*}{\Rightarrow} uAy$.

Since A is used recursively, the derivation can take the following form : $S \overset{*}{\Rightarrow} uAy \overset{*}{\Rightarrow} uvAxy$ and the final derivation should be of the form $S \overset{*}{\Rightarrow} uAy \overset{*}{\Rightarrow} uvAxy \overset{*}{\Rightarrow} uvwxy = z$ it implies that,

$A \overset{*}{\Rightarrow} vAx$ and $A \overset{*}{\Rightarrow} w$.

The longest string vwx is generated without recursion since it is assumed that $|vwx| \leq n$. Since CFG that generates CFL does not contain ε-productions or unit productions. It shows that every derivation step either increases the length of the sentential form or introduces a terminal. The derivation, $A \overset{*}{\Rightarrow} vAx$ shows that $|vx| \geq 1$.

The derivation, $S \overset{*}{\Rightarrow} uAy \overset{*}{\Rightarrow} uvAxy \overset{*}{\Rightarrow} uvwxy$, it implies that $A \overset{*}{\Rightarrow} vAx$ and $A \overset{*}{\Rightarrow} w$.

It follows that, $uv^i wx^i y \in L$ and hence the proof.

Applications of Pumping Lemma for CFL's

The general strategy used to prove that a given language is not context free is shown below:

1. Assume that the language L is infinite and it is context free.
2. Select the string z and break it into sub strings u, v, w, x, y such that $z = uvwxy$ where $|vwx| \leq n$ and $|vx| \geq 1$.
3. Find any i such that $uv^iwx^iy \notin L$. According to pumpping lemma, $uv^iwx^iy \in L$. So, the result is a contradiction to the assumption that the language is context free. Therefore, the given language L is not context free.

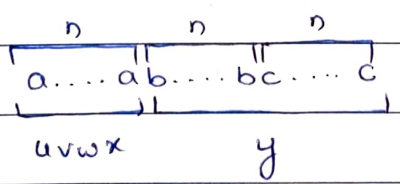
Q1. Show that $L = \{a^n b^n c^n \mid n \geq 0\}$ is not context free.

STEP 1 : Let L be context free and infinite.

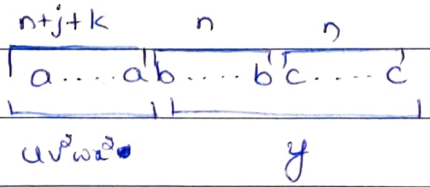
STEP 2 : Let $z = a^n b^n c^n \in L$, split z into uvwxy such that $|vwx| \leq n$ and $|vx| \geq 1$. and so $uv^iwx^iy \in L$ for $i = 0, 1, 2, \ldots$

CASE 1 : The string vwx is within a^n.

Let $v = a^j$, $x = a^k$ where $|vx| = j + k \geq 1$ and $|vwx| \leq n$ which can be shown pictorially as



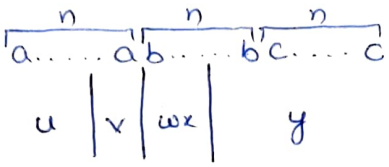
According to Pumping Lemma, $uv^2wx^2y \in L$ for $i = 2$ and the language generated is as shown below



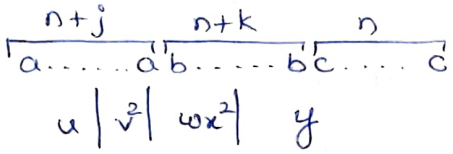
$uv^2wx^2y = a^{n+j+k}\, b^n c^n \notin L$ where $j+k \geq 1$, which is the contradiction.

CASE 2: The string vwx is in $a^n b^n$

Let $v = a^j$, $x = b^k$ where $|vx| = j+k \geq 1$ and $|vwx| \leq n$



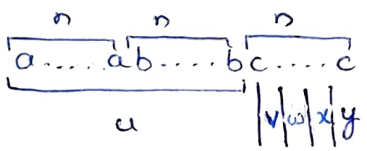
According to pumping lemma, $uv^2wx^2y \in L$ for $i = 2$ and the language generated is as shown below



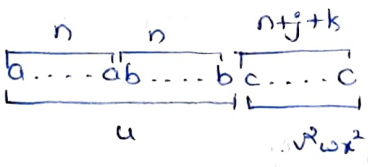
$uv^2wx^2y = a^{n+j}\, b^{n+k}\, c^n \notin L$ when $j+k \geq 1$, which is the contradiction.

CASE 3: The string vwx is within c^n.

Let $v = c^j$, $x = c^k$ where $|vx| = j+k \geq 1$ and $|vwx| \leq n$



According to pumping lemma, $uv^2wx^2y \in L$ for $i = 2$ and the language generated is,



$uv^2wx^2y = a^n b^n c^{n+j+k} \notin L$ when $j+k \geq 1$, which is the contradiction.

According to Pumping lemma, n number of a's should be followed by n number of b's which in turn should be followed by n number of c's. In all the three cases which is contradiction to the assumption that the language is context free.

So, the language $L = \{a^n b^n c^n \mid n \geq 0\}$ is not context free.

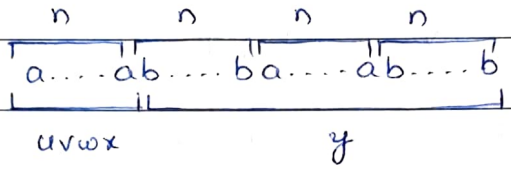
Q2. Show that $L = \{ww \mid w \in \{a,b\}^*\}$ is not context free.

STEP 1 : Assume that L is context free and is infinite.

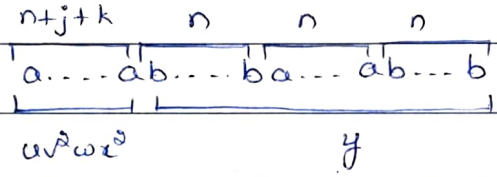
STEP 2 : Let $z = a^n b^n a^n b^n \in L$, split z into u, v, w, x, y such that $|vwx| \leq n$ and $|vx| \geq 1$, and $uv^i wx^i y \in L$ for $i = 0, 1, 2, \ldots$

CASE 1 : The string vwx is within first a^n

Let $v = a^j$, $x = a^k$ where $|vx| = j+k \geq 1$ and $|vwx| <= n$



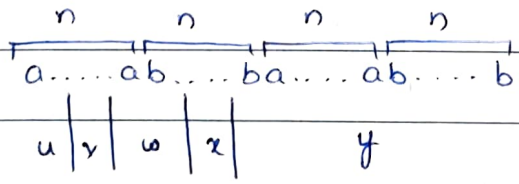
According to pumping lemma, $uv^2wx^2y \in L$ for $i = 2$.



$a^{n+j+k} b^n a^n b^n = uv^2wx^2y \notin L$ when $j+k \geq 1$, which is the contradiction.

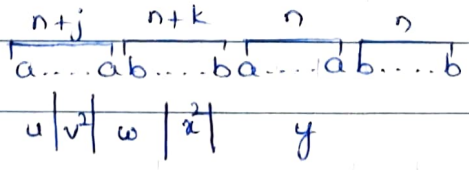
CASE 2 : Let v is in first a^n and x is in first b^n.

Let $v = a^j$, $x = b^k$ such that $|vx| = j+k \geq 1$ and $|vwx| \leq n$



(w has a's and b's)

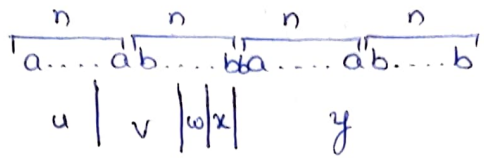
According to pumping lemma, $uv^2wx^2y \in L$ for $i = 2$.



$a^{n+j} b^{n+k} a^n b^n = uv^2wx^2y \notin L$ when $j+k \geq 1$, which is contradiction.

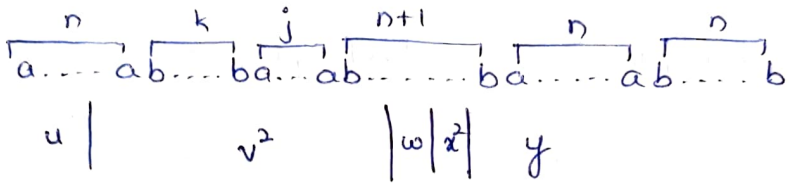
CASE 3 : Let v overlaps the first $a^n b^n$ and x is the first b^n.

Let $v = a^j b^k$ and $x = b^l$ such that $|vx| = j+k \geq 1$ and $|wx| \leq n$



(v has a's and b's
w and x has b's
y has b's followed by $a^n b^n$)

According to pumping lemma, $uv^i wx^i y \in L$ for $i = 2$.



$uv^2wx^2y \notin L$ when $j+k+1 \geq 1$. which is contradiction.

In all the three cases which is contradiction to the assumption that the language is context free.

So, the language $L = \{ww \mid w \in \{a,b\}^*\}$ is not context free.

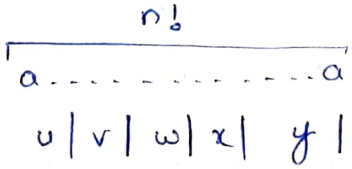
Q3. Show that $L = \{a^{n!} \mid n \geq 0\}$ is not context free.

STEP 1 : Assume that L is context free and is infinite.

STEP 2 : Let $z = a^{n!} \in L$ where $|a^{n!}| > n$.

Split $z = a^{n!}$ into u, v, w, x and y such that $|wx| \leq n$ and $|vx| \geq 1$.

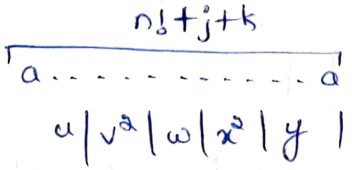
so that $uv^i wx^i y \in L$ for $i = 0, 1, 2, \ldots$



STEP 3: Let $v = a^j$, $x = a^k$ such that $|vx| = j+k \geq 1$ and $|wx| \leq n$.

According to pumping lemma

$n! = n! - 1$



when $n = 2$, $n! + j + k \leq n! + n$

$< n! + n!n$

$< n!(n+1)$

$< (n+1)!$

$n! < n! + j + k < (n+1)!$

$uv^2wx^2y = a^{n!+j+k}$ whenever $j+k \geq 1$, which is contradiction.

So, the language $L = \{a^{n!} \mid n \geq 0\}$ is not context free.

Q4. Show that $L = \{a^p b^q \mid p = q^2\}$ is not context free.

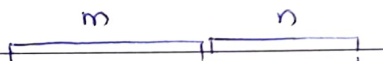
STEP 1 : Assume that L is context free and is infinite.

STEP 2 : Let $z = a^m b^n \in L$ where $m = n^2$ and $|a^m b^n| > n$.

Split the string $z = uvwxy = a^m b^n$ such that $|vwx| \leq n$ and $|vx| \geq 1$. so that $uv^i wx^i y \in L$ for $i = 0, 1, 2, \ldots$

STEP 3 : Assume that the string vwx is within $a^m b^n$.

Let $v = a^j$, $x = b^k$ such that $|vx| = j + k \geq 1$.



u | v | w | x | y

when $j + k \geq 1$ and $i = 0$, then, $uv^i wx^i y$ is,



u | w | y

$(n-k)^2 \leq (n-1)^2$ i.e., $a^{m-j} b^{n-k} = uwy$ and $j \neq 0$ and $k \neq 0$

$= n^2 - 2n + 1$ $\quad m - j \neq (n-k)^2$

$= m - 2n + 1$

$< m - j$ It means that $m - j$ is not a perfect square. According to Pumping lemma it should be a perfect square. So, $a^{m-j} b^{n-k} = uwy \notin L$ which is contradiction. So, the language $L = \{a^p b^q \mid p = q^2\}$ is not context free.

Q5. Show that $L = \{x \in \{0,1\}^* \mid |x| \text{ is a perfect square}\}$ is not context free.

STEP 1 : Assume that L is context free and is infinite.

STEP 2 : Let $z = 0^{n^2}$ where $|0^{n^2}| > n$.

split $z = 0^{n^2}$ into u, v, w, x, y such that $|vwx| \leq n$ and $|vx| \geq 1$.

so that $uv^i wx^i y \in L$ for $i = 0, 1, 2, \ldots$

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STEP 3 : $\underbrace{0 \ldots\ldots\ldots 0}_{n^2+j+k}$ Let $v = 0^j$, $x = 0^k$ such that $|vx| = j+k \geq 1$ and $|vwx| \leq n$.

$$u\,|\,v\,|\,w\,|\,x\,|\,y$$

when $n = 2$, $n^2 + j + k \leq n^2 + n \leq (n+1)^2$

i.e., $n^2 < (n+1)^2$

and hence its length is not a perfect square and therefore the string is not in L. i.e., $uv^2wx^2y \notin L$, which is contradiction.

So, the language $L = \{x \in \{0,1\}^* \mid |x| \text{ is a perfect square}\}$ is not context free.

CFLs are Closed Under Union, Concatenation and Star

Theorem :

If L_1 and L_2 are CFLs, then $L_1 \cup L_2$, $L_1 \cdot L_2$ and L_1^* also denote the CFLs and so the context free languages are closed under union, concatenation and, star-closure.

Proof : Let L_1 and L_2 are two CFLs generated by the CFGs. $G_1 = (V_1, T_1, P_1, S_1)$ and $G_2 = (V_2, T_2, P_2, S_2)$ respectively and assume that V_1 and V_2 are disjoint.

Case1 : Union of two CFLs is CFL.

let the language L_3 generated by the grammar

$G_3 = (V_1 \cup V_2 \cup S_3, T_1 \cup T_2, P_3, S_3)$

where, S_3 is a start symbol for the grammar G_3 and $S_3 \notin (V_1 \cup V_2)$

$P_3 = P_1 \cup P_2 \cup \{S_3 \to S_1 \mid S_2\}$

The grammar G_3 is context free and the language generated by this grammar is context free. So,

$L_3 = L_1 \cup L_2$

assume $w \in L_1$, then the possible derivation from S_3 is

$S_3 \Rightarrow S_1 \overset{*}{\Rightarrow} w$

if $w \in L_2$, then $S_3 \Rightarrow S_2 \overset{*}{\Rightarrow} w$

So, if $w \in L_3$, one of the derivations $S_3 \Rightarrow S_1$ or $S_3 \Rightarrow S_2$ is possible.

In the first case, all the variables in V_1 and all the terminals in T_1 may be used to get the derivation $S_1 \overset{*}{\Rightarrow} w$ which uses only the productions in P_1.

Similarly all the variables in V_2 and all the terminals in T_2 may be used to get the derivation $S_2 \overset{*}{\Rightarrow} w$ which uses only the productions in P_2 and it follows that $L_3 = L_1 \cup L_2$, Thus, the context free languages are closed under union.

Case 2: Concatenation of two CFLs is CFL.

Let the language L_4 generated by the grammar

$G_4 = (V_1 \cup V_2 \cup S_4, T_1 \cup T_2, P_4, S_4)$

where

S_4 is the start symbol for the grammar G_4 and $S_4 \notin (V_1 \cup V_2)$

$P_4 = P_1 \cup P_2 \cup \{ S_4 \rightarrow S_1 S_2 \}$

The grammar G_4 is context free and the language generated by this grammar is context free and so $L_3 = L_1 L_2$

Thus, the context free languages are closed under concatenation

Case 3: CFLs are closed under star-closure.

Let the language L_5 generated by the grammar

$G_5 = (V_1 \cup S_5, T_1, P_5, S_5)$

where

S_5 is the start symbol for the grammar G_5

$P_5 = P_1 \cup \{ S_5 \rightarrow S_1 S_5 | \varepsilon \}$

The grammar G_5 is context free and the language generated by this grammar is context free and so $L_5 = L_5^*$.

Thus, the context free languages are closed under star-closure.

Thus, the context free languages are closed under union, concatenation and star- "

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CFLs are Not Closed Under Intersection

Theorem : The CFLs are not closed under intersection. If L_1 and L_2 are context free languages, it is not always true that $L_1 \cap L_2$ is context free language.

Proof : Consider the two languages

$L_1 = \{ a^n b^n c^m \mid n \geq 0, m \geq 0 \}$

and

$L_2 = \{ a^n b^m c^m \mid n \geq 0, m \geq 0 \}$

The two languages are context free, corresponding context-free grammars are

$S \rightarrow S_1 S_2$

$S_1 \rightarrow a S_1 b \mid \varepsilon$

$S_2 \rightarrow c S_2 \mid \varepsilon$

and

$S \rightarrow aS \mid S_1 \mid \varepsilon$

$S_1 \rightarrow bS_1 c \mid \varepsilon$

or

$S \rightarrow S_1 S_2$

$S_1 \rightarrow aS_1 \mid \varepsilon$

$S_2 \rightarrow bS_2 c \mid \varepsilon$

So, $L_1 \cap L_2 = \{ a^n b^n c^n \mid n \geq 0 \}$

Thus, the family of context free languages is not closed under intersection.

CFLs are Not Closed Under Complementation

Theorem : The CFLs are not closed under complementation. If L is context free language, it is not true that complement of L is context free language.

Proof : Suppose that context-free languages are closed under complementation. So, if L_1 and L_2 are context-free languages, then $\bar{L}_1$ and $\bar{L}_2$ are also context free. CFLs are closed under union. So, $\bar{L}_1 \cup \bar{L}_2$ must be context free.

Since its assumed that the CFLs are closed under complementation, $\bar{L}_1 \cup \bar{L}_2$ must be context free.

But, according to de Morgan's Law

$\bar{L}_1 \cup \bar{L}_2 = L_1 \cap L_2$ So, $L_1 \cap L_2$ must be context free which is a contradiction.

Since the context free languages are not closed under intersection, the assumption that the CFLs are closed under complementation is not true. So, the family of context free languages are closed under complementation.